

DIGITAL SAT MATH PRACTICE

Radical and Rational Exponents — SOLUTIONS

COMPLETE SOLUTIONS WITH TRAP ALERTS AND STRATEGY

Question 1 [Easy]

Correct Answer: A

Step by Step:

Step 1: $\sqrt{(49x^4)} = \sqrt{49} \cdot \sqrt{(x^4)}$.

Step 2: $\sqrt{49} = 7$ and $\sqrt{(x^4)} = x^2$ (since $x^4 = (x^2)^2$).

Step 3: Result: $7x^2$.

Why Each Wrong Option Fails:

Option B: Keeps the exponent 4 unchanged. The square root halves the exponent.

Option C: Doesn't take the square root of 49.

Option D: Divides the exponent by 4 instead of 2.

■ **TRAP ALERT:** The square root halves the exponent: $\sqrt{(x^4)} = x^2$, not x or x^4 .

■ **KEY INSIGHT:** Square root rule: $\sqrt{(x^n)} = x^{(n/2)}$. So $\sqrt{(x^4)} = x^{(4/2)} = x^2$.

■ **FAST METHOD:** Split: $\sqrt{49} = 7$, exponent $4 \div 2 = 2$. Answer: $7x^2$.

■ **VERDICT:** Hand solve is faster. Direct application of the square root exponent rule.

Question 2 [Easy]

Correct Answer: A

Step by Step:

Step 1: $27^{(1/3)}$ means the cube root of 27.

Step 2: Since $3^3 = 27$, the cube root of 27 is 3.

Why Each Wrong Option Fails:

Option B: $9 = 27^{(2/3)}$. This squares after taking the cube root.

Option C: $81 = 27 \times 3$. This multiplies instead of finding the root.

Option D: $1/3$ confuses the exponent with the answer.

■ **TRAP ALERT:** Students sometimes confuse $x^{(1/3)}$ with $x \times (1/3) = x/3$.

■ **KEY INSIGHT:** $x^{(1/n)}$ = the n th root of x . The denominator tells you which root.

■ **FAST METHOD:** Ask: what cubed equals 27? Answer: 3.

■ **VERDICT:** Hand solve is faster. Recognizing perfect cubes is instant.

Question 3 [Easy]

Correct Answer: A

Step by Step:

Step 1: $x^{(3/4)}$ means the 4th root of x^3 , which is written as $\sqrt[4]{x^3}$.

Step 2: In the form $x^{(a/b)}$: the numerator a is the power, the denominator b is the root.

Why Each Wrong Option Fails:

Option B: $\sqrt[3]{x^4} = x^{(4/3)}$. The numerator and denominator are flipped.

Option C: $(x^3)/4$ treats the fraction as division, not an exponent.

Option D: $3\sqrt{x} = 3x^{(1/2)}$. The 3 is a coefficient, not an exponent.

■ **TRAP ALERT: Choice B flips the fraction: denominator = root, numerator = power. Don't mix them up.**

■ **KEY INSIGHT: $x^{(a/b)} = \sqrt[b]{x^a}$. Denominator = root (outside), Numerator = power (inside).**

■ **FAST METHOD: 3/4: the 4 goes outside the radical, the 3 stays as the power of x inside.**

■ **VERDICT: Hand solve is faster. Pure notation recognition, no computation.**

Question 4 [Easy]

Correct Answer: B

Step by Step:

Step 1: $16^{(3/4)} = (\sqrt[4]{16})^3$.

Step 2: $\sqrt[4]{16} = 2$ (since $2^4 = 16$).

Step 3: $2^3 = 8$.

Why Each Wrong Option Fails:

Option A: $4 = 16^{(1/2)}$. This takes the square root instead of the 4th root, then forgets to cube.

Option C: $12 = 16 \times (3/4)$. This multiplies instead of exponentiating.

Option D: $64 = 16^{(1/4)}$ done incorrectly; cubing 16 first gives 4096, and $\sqrt[4]{4096} = 8$.

■ **TRAP ALERT: Order of operations: take the root first, THEN raise to the power. Root first avoids huge numbers.**

■ **KEY INSIGHT: Always root first when evaluating $x^{(a/b)}$: find $\sqrt[b]{x}$ first, then raise to power a .**

■ **FAST METHOD: $\sqrt[4]{16} = 2$, then $2^3 = 8$. Two quick mental steps.**

■ **VERDICT: Hand solve is faster. Recognizing $16 = 2^4$ makes this trivial.**

Question 5 [Medium]

Correct Answer: B

Step by Step:

Step 1: Square both sides: $5k + 11 = (k + 1)^2 = k^2 + 2k + 1$.

Step 2: Rearrange: $k^2 + 2k + 1 - 5k - 11 = 0 \rightarrow k^2 - 3k - 10 = 0$.

Step 3: Factor: $(k - 5)(k + 2) = 0$, so $k = 5$ or $k = -2$.

Step 4: Check $k = 5$: $\sqrt{(25 + 11)} = \sqrt{36} = 6$ and $5 + 1 = 6$. ✓

Step 5: Check $k = -2$: $\sqrt{(-10 + 11)} = \sqrt{1} = 1$ and $-2 + 1 = -1$. $1 \neq -1$. ✗ (Extraneous!)

Step 6: Only $k = 5$ is valid.

Why Each Wrong Option Fails:

Option A: $k = -2$ fails the check. $\sqrt{1} = 1$ but $-2 + 1 = -1$.

Option C: $k = -2$ is extraneous; only $k = 5$ works.

Option D: There is one valid solution: $k = 5$.

■ **TRAP ALERT:** Squaring both sides can introduce extraneous solutions. ALWAYS check answers in the original equation.

■ **KEY INSIGHT:** Extraneous solutions appear when you square both sides. The check step is mandatory.

■ **FAST METHOD:** Factor the quadratic, then plug both answers back into the ORIGINAL equation (not the squared version).

■  **VERDICT:** Desmos is helpful. Graph $y = \sqrt[3]{5x+11}$ and $y = x+1$ to see where they intersect.

Question 6 [Medium]

Correct Answer: A

Step by Step:

Step 1: Apply exponent to each factor: $8^{(2/3)} \cdot (m\blacksquare)^{(2/3)}$.

Step 2: $8^{(2/3)} = (\sqrt[3]{8})^2 = 2^2 = 4$.

Step 3: $(m\blacksquare)^{(2/3)} = m^{(6 \times 2/3)} = m\blacksquare$.

Step 4: Result: $4m\blacksquare$.

Why Each Wrong Option Fails:

Option B: Uses $6 \times (3/2) = 9$ for the exponent (flipped the fraction).

Option C: Gets the coefficient wrong: $\sqrt[3]{8} = 2$, but forgets to square it.

Option D: Both errors combined: wrong coefficient AND flipped exponent fraction.

■ **TRAP ALERT:** Choice C forgets to apply the numerator (2) to the cube root result. $8^{(2/3)} = 4$, not 2.

■ **KEY INSIGHT:** Power of a power rule: $(x^a)^b = x^{(ab)}$. Multiply exponents.

■ **FAST METHOD:** $\sqrt[3]{8} = 2$, squared = 4. Exponent: $6 \times 2/3 = 4$. Answer: $4m\blacksquare$.

■  **VERDICT:** Hand solve is faster. Straightforward exponent arithmetic.

Question 7 [Medium]

Correct Answer: D

Step by Step:

Step 1: $x^{(3/2)} = (x^{(1/2)})^3$.

Step 2: Since $x^{(1/2)} = 5$, we get $5^3 = 125$.

Why Each Wrong Option Fails:

Option A: $15 = 5 \times 3$. Multiplies the base by the numerator instead of cubing.

Option B: $25 = 5^2$. Squares instead of cubing.

Option C: $75 = 25 \times 3$. Squares then multiplies by 3.

■ **TRAP ALERT:** Choice A (15) is the most common trap: students multiply 5×3 instead of computing 5^3 .

■ **KEY INSIGHT:** $x^{(3/2)} = (x^{(1/2)})^3$. Use the known value and raise it to the numerator power.

■ **FAST METHOD:** $x^{(1/2)} = 5$, so $x^{(3/2)} = 5^3 = 125$.

■  **VERDICT:** Hand solve is faster. One step of cubing.

Question 8 [Medium]

Correct Answer: B

Step by Step:

Step 1: When multiplying powers with the same base, add exponents.

Step 2: $x^{(2/5)} \cdot x^{(3/5)} = x^{(2/5 + 3/5)} = x^{(5/5)} = x^1 = x$.

Why Each Wrong Option Fails:

Option A: $x^{(6/25)}$ multiplies the exponents ($2/5 \times 3/5$). That rule is for power of a power, not multiplication.

Option C: $x^{(6/5)}$ would be correct if the exponents were $3/5 + 3/5$, not $2/5 + 3/5$.

Option D: $x^{(5/10)} = x^{(1/2)}$. This adds numerators and denominators separately, which is incorrect fraction addition.

■ **TRAP ALERT:** Choice A multiplies exponents instead of adding. Multiplication of bases → ADD exponents.

■ **KEY INSIGHT:** $x^a \cdot x^b = x^{(a+b)}$. Add exponents when bases are the same and you're multiplying.

■ **FAST METHOD:** $2/5 + 3/5 = 5/5 = 1$. Answer: x .

■ **VERDICT:** Hand solve is faster. Simple fraction addition.

Question 9 [Medium]

Correct Answer: A

Step by Step:

Step 1: $\sqrt[3]{64y^9} = \sqrt[3]{64} \cdot \sqrt[3]{y^9}$.

Step 2: $\sqrt[3]{64} = 4$ (since $4^3 = 64$).

Step 3: $\sqrt[3]{y^9} = y^{(9/3)} = y^3$.

Step 4: Result: $4y^3$.

Why Each Wrong Option Fails:

Option B: $8 = \sqrt{64}$. Uses square root instead of cube root for 64.

Option C: y^3 divides exponent by 1.5 instead of 3.

Option D: Both errors: wrong root for 64 and wrong exponent division.

■ **TRAP ALERT:** $\sqrt{64} = 8$, but $\sqrt[3]{64} = 4$. Make sure you use the correct root index.

■ **KEY INSIGHT:** For $\sqrt[n]{x^a} = x^{(a/n)}$. Divide the exponent by the root index.

■ **FAST METHOD:** Coefficient: $4^3 = 64$, so $\sqrt[3]{64} = 4$. Exponent: $9/3 = 3$.

■ **VERDICT:** Hand solve is faster. Recognize perfect cubes and divide exponents.

Question 10 [Medium]

Correct Answer: 4

Step by Step:

Step 1: $32^{(2/5)} = (\sqrt[5]{32})^2$.

Step 2: $\sqrt[5]{32} = 2$ (since $2^5 = 32$).

Step 3: $2^2 = 4$.

■ **TRAP ALERT:** Students may try to compute $32^2 = 1024$ first, then find the 5th root, making the arithmetic hard. Root first!

■ **KEY INSIGHT:** Always take the root first when the base is a perfect power: $32 = 2^5$.

■ **FAST METHOD:** $\sqrt[5]{32} = 2$, then $2^2 = 4$. Enter 4.

■ **VERDICT:** Hand solve is faster. Recognizing $32 = 2^5$ is the key step.

Question 11 [Medium]

Correct Answer: B

Step by Step:

Step 1: A negative exponent means reciprocal: $x^{-1/2} = 1/(x^{1/2})$.

Step 2: $x^{1/2} = \sqrt{x}$.

Step 3: So $x^{-1/2} = 1/\sqrt{x}$.

Why Each Wrong Option Fails:

Option A: $-\sqrt{x}$ treats the negative exponent as a negative sign in front.

Option C: $\sqrt{-1/x}$ places the negative inside the radical, which isn't valid for real numbers.

Option D: $-x/2$ treats the exponent as multiplication by $-1/2$.

■ **TRAP ALERT:** A negative exponent means "1 over," not "negative." $x^{-1/2} \neq -\sqrt{x}$.

■ **KEY INSIGHT:** $x^{-n} = 1/x^n$. The negative exponent creates a reciprocal.

■ **FAST METHOD:** Negative = flip. Half = square root. So $x^{-1/2} = 1/\sqrt{x}$.

■ **VERDICT:** Hand solve is faster. No computation, just exponent rule recognition.

Question 12 [Hard]

Correct Answer: B

Step by Step:

Step 1: $3^{(2x)} = 81 = 3^4$, so $2x = 4$, giving $x = 2$.

Step 2: $2^y = 16 = 2^4$, so $y = 4$.

Step 3: $x + y = 2 + 4 = 6$.

Why Each Wrong Option Fails:

Option A: Gets $x = 2$ but sets $y = 2$ (using $\sqrt{16} = 4$ then taking $\log 4 = 2$).

Option C: Sets $2x = 4$ so $x = 4$ (forgetting to divide by 2) and $y = 4$.

Option D: Gets $x = 4$ (doesn't divide $2x$) and $y = 6$ (wrong power of 2).

■ **TRAP ALERT:** $3^{(2x)} = 3^4$ gives $2x = 4$, not $x = 4$. Don't forget to divide by 2.

■ **KEY INSIGHT:** Match bases: rewrite the right side as a power of the same base as the left.

■ **FAST METHOD:** $81 = 3^4 \rightarrow 2x = 4 \rightarrow x = 2$. $16 = 2^4 \rightarrow y = 4$. Sum = 6.

■ **VERDICT:** Hand solve is faster. Pattern matching with powers, no calculator needed.

Question 13 [Hard]

Correct Answer: 5

Step by Step:

Step 1: $(x^a)^3 = x^{(3a)}$ by the power rule.

Step 2: $x^{(3a)} \cdot x^4 = x^{(3a + 4)}$ by the product rule.

Step 3: Set equal to x^{19} : $3a + 4 = 19$.

Step 4: $3a = 15$, so $a = 5$.

■ **TRAP ALERT:** Students may multiply $3 \times 4 = 12$ and add to a , mixing up the rules.

■ **KEY INSIGHT:** Power of a power: multiply exponents. Product of powers: add exponents. Apply in order.

■ **FAST METHOD:** $3a + 4 = 19 \rightarrow a = 5$. Simple linear equation.

■  **VERDICT:** Hand solve is faster. Set up and solve a one step equation.

Question 14 [Hard]

Correct Answer: A

Step by Step:

Step 1: Distribute the $1/2$ exponent: $4^{(1/2)} \cdot (x^2)^{(1/2)} \cdot (y^{-2})^{(1/2)}$.

Step 2: $4^{(1/2)} = 2$.

Step 3: $(x^2)^{(1/2)} = x$.

Step 4: $(y^{-2})^{(1/2)} = y^{-1} = 1/y$.

Step 5: Result: $2x^3/y$.

Why Each Wrong Option Fails:

Option B: Treats y^{-2} as y^{+2} and gets y^1 instead of y^{-1} .

Option C: Doesn't take the square root of 4.

Option D: Multiplies the x exponent by 2 instead of $1/2$.

■ **TRAP ALERT:** The negative exponent on y stays negative: $(-2)(1/2) = -1$, giving $1/y$.

■ **KEY INSIGHT:** Distribute the outer exponent to every factor, including negative exponents.

■ **FAST METHOD:** Coefficient: $\sqrt{4} = 2$. x : $6/2 = 3$. y : $-2/2 = -1$. Answer: $2x^3y^{-1} = 2x^3/y$.

■  **VERDICT:** Hand solve is faster. Apply the $1/2$ exponent to each part systematically.

Question 15 [Hard]

Correct Answer: A

Step by Step:

Step 1: $\sqrt{50x^3} = \sqrt{25 \cdot 2 \cdot x^2 \cdot x}$.

Step 2: $= \sqrt{25} \cdot \sqrt{x^2} \cdot \sqrt{2x}$.

Step 3: $= 5x\sqrt{2x}$.

Why Each Wrong Option Fails:

Option B: Doesn't take the square root of 25 (leaves it as 25 instead of 5).

Option C: Takes $\sqrt{x^3} = x^2$ instead of $x\sqrt{x}$. The exponent 3 is odd, so one x stays inside.

Option D: Uses $\sqrt{50} = 10$ (incorrect; $\sqrt{50} = 5\sqrt{2}$).

■ **TRAP ALERT:** $\sqrt{x^3} = x\sqrt{x}$, not x^2 or $x^{3/2}$. Split $x^3 = x^2 \cdot x$, then $\sqrt{x^2} = x$ comes out.

■ **KEY INSIGHT:** Factor out the largest perfect square from under the radical, both for numbers and variables.

■ **FAST METHOD:** $50 = 25 \cdot 2$, $x^3 = x^2 \cdot x$. Pull out $\sqrt{25} = 5$ and $\sqrt{x^2} = x$. Remainder: $\sqrt{2x}$.

■  **VERDICT:** Hand solve is faster. Factor and simplify, no calculator needed.

Question 16 [Hard]

Correct Answer: B

Step by Step:

Step 1: First find $g(64) = 64^{(1/3)} = 4$ (since $4^3 = 64$).

Step 2: Then $f(4) = 2(4)^{3/2}$.

Step 3: $4^{3/2} = (\sqrt{4})^3 = 2^3 = 8$.

Step 4: $f(4) = 2 \times 8 = 16$.

Why Each Wrong Option Fails:

Option A: $8 = 4^{3/2}$ without multiplying by the coefficient 2.

Option C: $32 = 2 \times 16$, which would result from computing 4^2 instead of $4^{3/2}$.

Option D: 64 is the original input, not the output. This is a trap for students who skip computation.

■ **TRAP ALERT:** Don't forget the coefficient 2 in $f(x) = 2x^{3/2}$. It's easy to compute $4^{3/2} = 8$ and stop.

■ **KEY INSIGHT:** Composition: work inside out. Compute $g(4)$ first, then plug the result into f .

■ **FAST METHOD:** $g(4) = 2$. Then $4^{3/2} = 8$. Then $2 \times 8 = 16$.

■ **VERDICT:** Hand solve is faster. Chain of simple computations with perfect powers.

Question 17 [Hard]

Correct Answer: 15

Step by Step:

Step 1: For x : $x^a/x^{-2} = x^{4-(-2)} = x^6$. So $a = 6$.

Step 2: For y : $y^{-6}/y^3 = y^{-6-3} = y^{-9}$. So $b = -9$.

Step 3: $a - b = 6 - (-9) = 6 + 9 = 15$.

■ **TRAP ALERT:** Subtracting a negative: $a - b = 6 - (-9)$. Students often get $6 - 9 = -3$ by dropping the negative.

■ **KEY INSIGHT:** Quotient rule: $x^a / x^b = x^{a-b}$. Be careful with negative exponents.

■ **FAST METHOD:** x : $4 - (-2) = 6$. y : $-6 - 3 = -9$. Answer: $6 - (-9) = 15$.

■ **VERDICT:** Hand solve is faster. Integer subtraction with signs. Enter 15.

Question 18 [Hard]

Correct Answer: D

Step by Step:

Step 1: $3 \cdot p^{1/3} = 12$.

Step 2: $p^{1/3} = 4$.

Step 3: Cube both sides: $p = 4^3 = 64$.

Why Each Wrong Option Fails:

Option A: $4 = p^{1/3}$. This is the cube root, not p itself.

Option B: $36 = 12 \times 3$. Multiplies instead of cubing.

Option C: $48 = 12 \times 4$. Another multiplication error.

■ **TRAP ALERT:** Choice A (4) is the value of $p^{1/3}$, NOT p . You must cube to find p .

■ **KEY INSIGHT:** If $p^{1/3} = 4$, then $p = 4^3 = 64$. The inverse of a cube root is cubing.

■ **FAST METHOD:** $3p^{1/3} = 12 \rightarrow p^{1/3} = 4 \rightarrow p = 64$.

■ **VERDICT:** Hand solve is faster. Three step algebra, no calculator needed.

Question 19 [Hard]

Correct Answer: 4

Step by Step:

Step 1: Add exponents: $5^x \cdot 5^{(x+1)} = 5^{(x + x + 1)} = 5^{(2x + 1)}$.

Step 2: Set equal: $2x + 1 = 9$.

Step 3: $2x = 8$, so $x = 4$.

■ **TRAP ALERT:** Students may multiply the exponents $(x)(x+1)$ instead of adding them.

■ **KEY INSIGHT:** When multiplying same base: **ADD** exponents. $x + (x + 1) = 2x + 1$.

■ **FAST METHOD:** $2x + 1 = 9 \rightarrow x = 4$. Enter 4.

■  **VERDICT:** Hand solve is faster. Simple exponent rule then linear equation.

Question 20 [Hard]

Correct Answer: B

Step by Step:

Step 1: Rewrite in exponent form: $(a^2b)^{(1/3)} \cdot (ab^{\blacksquare})^{(1/3)}$.

Step 2: Combine under one cube root: $(a^2b \cdot ab^{\blacksquare})^{(1/3)} = (a^3b^{\blacksquare})^{(1/3)}$.

Step 3: $(a^3)^{(1/3)} = a$ and $(b^{\blacksquare})^{(1/3)} = b^2$.

Step 4: Result: ab^2 , so $m = 1$ and $n = 2$.

Step 5: $m + n = 1 + 2 = 3$.

Why Each Wrong Option Fails:

Option A: Gets $a^{(2/3)} \cdot a^{(1/3)} = a$ correctly but miscalculates b .

Option C: $7/3$ would come from not simplifying fully and adding $2/3 + 5/3$ for b .

Option D: $8/3$ comes from adding all the individual exponents without combining.

■ **TRAP ALERT:** Students may try to distribute the cube root separately and lose track of the exponents.

■ **KEY INSIGHT:** When the root index is the same, combine the expressions under one radical first, then simplify.

■ **FAST METHOD:** $a: 2+1 = 3$, divided by $3 = 1$. $b: 1+5 = 6$, divided by $3 = 2$. $m+n = 1+2 = 3$.

■  **VERDICT:** Hand solve is faster. Add exponents under the radical, then divide each by 3.

TOPIC SUMMARY: RADICAL AND RATIONAL EXPONENTS

- $x^{(1/n)}$ = the n th root of x . The denominator of a rational exponent is the root index.
- $x^{(a/b)} = (\sqrt[b]{x})^a$. Denominator = root, Numerator = power. Always take the root first for cleaner arithmetic.
- Product Rule: $x^a \cdot x^b = x^{(a+b)}$. ADD exponents when multiplying same bases.
- Quotient Rule: $x^a / x^b = x^{(a-b)}$. SUBTRACT exponents when dividing same bases.
- Power Rule: $(x^a)^b = x^{(ab)}$. MULTIPLY exponents for power of a power.
- Negative Exponents: $x^{(-n)} = 1/x^n$. Negative means reciprocal, NOT negative value.
- Simplifying Radicals: Factor out the largest perfect square (or cube, etc.) from under the radical.
- Extraneous Solutions: When you square both sides of an equation, always check solutions in the original.
- Common Perfect Powers to Know: $2^4=16$, $2^5=32$, $3^3=27$, $4^3=64$, $5^3=125$, $2^{10}=1024$.
- Most rational exponent problems on the DSAT are faster by hand. Desmos helps only for radical equations with extraneous solution checks.